Class 4, Week 11: Features and choosing a model, and your project

Lecturer guide for CML3112, CML3113 and CML3111. Keep it private.

The class in one paragraph

Which model should we recommend for predicting cooling load, and how do we explain the choice to someone who pays?

Welcome to the last teaching class of the semester. Over three weeks you have built a model, judged it, and learned how to trust it. Today you compare several models and choose one. In real work you will have several candidate models for predicting cooling load, and you must choose one fairly and explain the choice to the client who pays for the cooling plant. The ideas come first, then the lab, then the quiz for this block.

When each cohort meets

Cohort	Programmes	Date	Time and room
CML3112	Civil Engineering and Architecture	Tuesday 24 November 2026	11:00 to 14:00, Room 106
CML3113	Mining and Petroleum Engineering	Friday 27 November 2026	08:00 to 11:00, Room 105
CML3111	Electrical, Mechatronics, Robotics and Communications	Friday 27 November 2026	11:00 to 14:00, Room 115

Before the class

- Open the deck in Slide Show mode and the lab notebook in Colab before anyone arrives. Run the notebook from the top once so you know it works on your account.
- Put the lab notebook and engineering_data.csv in the shared class folder and post the link. Every learner needs both files in Colab, in the Files panel.
- Write the "check your numbers" table on the board before the lab (numbers for your cohort are in the table below).
- Prepare the Block 2 quiz for the end of the class and confirm in writing how long it stays open. The deck says it opens in the last 20 minutes and stays open for 24 hours. The 25-minute slot on the clock is for opening it and explaining it. The deck has 32 authored slides, so keep the formula slides brisk.
- Have the rehearsal (the practice version of the project) ready to release right after the class. The project is an individual, timed, auto-graded 3-hour sitting in the Week 12 class slot, covering Weeks 1 to 11, worth 45 percent (Part A 30, Part B 15). The deck gives the proposed times as lecturer to confirm (CML3112 Tue 1 Dec 11:00 to 14:00, CML3113 Fri 4 Dec 08:00 to 11:00, CML3111 Fri 4 Dec 11:00 to 14:00). Confirm them in writing.

The clock

Tuesday cohort (CML3112): the first full teaching of this class. Friday cohorts (CML3113, CML3111): the same deck and the same clock, because from Class 2 on all three cohorts are on the same footing.

CML3112, Tuesday

Starts	Minutes	What happens
0:00	10	Recall from Class 3
0:10	35	Concepts, part 1: features, scaling and two new models
0:45	30	Concepts, part 2: a fair comparison, the memo and the project

Starts	Minutes	What happens
1:15	15	Break
1:30	55	Lab: the model comparison table, in the notebook
2:25	25	Open and explain the Block 2 quiz (open 24 hours)
2:50	10	Close and the project briefing

CML3113 and CML3111, Friday

Starts	Minutes	What happens
0:00	10	Recall from Class 3
0:10	35	Concepts, part 1: features, scaling and two new models
0:45	30	Concepts, part 2: a fair comparison, the memo and the project
1:15	15	Break
1:30	55	Lab: the model comparison table, in the notebook
2:25	25	Open and explain the Block 2 quiz (open 24 hours)
2:50	10	Close and the project briefing

Slide map

The deck has two halves. Slides up to the divider are concepts: no laptops. After the divider come the lab slides and the notebook. You switch once.

Slide	Title	Part	Friday recap
1	Features and choosing a model, and your project	Concepts	
2	Today in one view: what you will leave with	Concepts	
3	Your three hours: models and memo first, then the lab	Concepts	
4	Three questions from last time	Concepts	
5	Words we use today	Concepts	
6	Class 3 in short: overfitting, cross-validation and data leakage	Concepts	
7	In real work you compare several models, choose one and explain why	Concepts	
8	A new feature can carry what you know about the machine	Concepts	
9	The new feature is a product, and the extended linear regression model gets one more weight	Concepts	
10	A feature ablation tests one change on its own, with everything else kept the same	Concepts	
11	Removing ambient_c hurts a lot, and the new features do not help	Concepts	
12	One-hot encoding turns a categorical feature into 0 and 1 columns, and equipment_id stays out of the features	Concepts	
13	Scaling puts every feature on the same ruler	Concepts	
14	Ridge regression adds an L2 penalty on large weights so a flexible curve stops fitting noise	Concepts	
15	Ridge regression minimises squared errors plus alpha times squared weights	Concepts	

Slide	Title	Part	Friday recap
16	A bigger alpha shrinks the weights, but on this ridge regression on two features the error grows	Concepts	
17	k-nearest neighbours (k-NN) predicts from the k training rows most like the new row	Concepts	
18	k-NN measures distance on the scaled features, then averages the k nearest targets	Concepts	
19	Worked example: the four closest training rows to the new row, record 5	Concepts	
20	The size of k matters: too small chases noise, too large flattens the pattern	Concepts	
21	A fair model comparison keeps four things the same for every model	Concepts	
22	Every model gets MAE, RMSE and four-fold cross-validation scores	Concepts	
23	Model comparison table: five candidate models on the same 15 validation rows	Concepts	
24	Model selection: choose the simplest model that is good enough (parsimony)	Concepts	
25	The model comparison table has three limits, and you must say them out loud	Concepts	
26	The held-out test set is used once, after the model is chosen	Concepts	
27	A decision memo fits on one page and has seven parts	Concepts	
28	Your project is an individual, timed practical sitting that the computer grades	Concepts	
29	The project is auto-graded and worth 45 percent of the course	Concepts	
30	A rehearsal comes first, and the sitting is in your Week 12 class slot	Concepts	
31	Read the recipe before you write it: a dictionary of models and a loop	Concepts	
32	Today in five lines: test ideas, compare fairly, choose simply, write it down	Concepts	
33	Part 2: the lab	Lab	
34	The lab has nine notebook steps in five groups, and the model comparison table is the centre	Lab	
35	Check your numbers against these before you move on	Lab	
36	Four errors you will meet, what they mean and what to do	Lab	
37	How this class counts towards your coursework	Lab	
38	Save the notebook and keep your memo for the project	Lab	

Questions to ask, and a good answer

Ask	A good answer
What is a feature ablation?	Change one thing, keep everything else the same, and compare on the same validation rows.
What is an interaction feature?	A column made by multiplying two features, here load_pct times ambient_c.
Why do we leave equipment_id out of the features?	The model must work on a new machine. A group it never saw would have no column of its own.
Why do ridge regression and k-NN need scaled features?	The penalty and the distance both depend on units, so a feature with big numbers would dominate.

Ask	A good answer
What does a bigger alpha do?	Alpha is the regularisation strength. A bigger alpha makes the L2 penalty stronger and the curve flatter. Too big and the model underfits.
Write the ridge regression loss in words.	The sum of the squared errors on the training rows, plus alpha times the sum of the squared weights.
How does k-NN measure closeness?	By the Euclidean distance on the scaled features: square the gaps, add them, take the square root.
What does k mean in k-NN?	The number of closest training rows that are averaged to make the prediction.
Why does the table show a cross-validated range?	One validation group of 15 rows can mislead. The range shows how much the score moves when other machines take a turn.
When two models score about the same, which do you pick?	The simpler one (parsimony), because it is easier to explain, check and maintain.
Who may open the test set, and when?	Nobody until the model is chosen, and then only once. Otherwise it stops being a fair held-out test.

Ideas learners often get wrong

What a learner may say	What to say back
A clever new feature must help	Test it. Here the load_x_ambient column changes the error by hardly anything.
The model with the lowest number on the table is the one to choose	Look at the cross-validated range too, and prefer the simpler model when scores are close.
Equipment_id should be a feature because it is in the file	The model must work on a machine it has never met, and a new machine has no column of its own.
A bigger alpha or a bigger k is always safer	Too much penalty or too many neighbours flattens the pattern. The error goes up again.
The test set can be used to pick between two close models	It is opened once, at the very end. After that it cannot give a fair number.

Walking the lab

Step	What to look for
Step 1: prepare	Rows left to model with: 88. Rows 58, 15, 15. Anyone with different numbers has run a cell twice or changed a cleaning rule.
Step 2: first loop	Three models, one loop (pipeline with median imputation). Baseline model, linear regression and decision tree match the Class 1 model comparison table. Ask why the loop keeps the comparison fair.
Step 3: features	Four versions of the linear regression model. Removing ambient_c hurts a lot. The product and the squared load change the MAE by less than 0.03. Resist the urge to explain the product away: the lesson is that the data decide.
Step 4: one-hot encoding	The validation group has an is_E5 column that the training table lacks. This is the whole argument for leaving equipment_id out.
Step 5: scaling and ridge regression	Ridge regression on two features gets slightly worse as alpha grows. The degree-6 curve on load alone improves at alpha 1 and worsens at alpha 100, and its biggest weight shrinks from thousands to single figures.
Step 6: k-NN	k = 3 is best of 1, 3 and 20. The three neighbours of the first validation row are printed, and their average is the prediction.

Step	What to look for
Step 7: the model comparison table	Five rows. Linear regression has the lowest cross-validated MAE. In one cohort a different model has a lower validation MAE: use it to talk about 15 rows and luck.
Step 8 and 9: memo and save	Collect one memo sentence from three learners. Insist on restart and run all before saving.

The numbers each cohort should see

Check	CML3112 (kW)	CML3113 (kWh/t)	CML3111 (deg C)
Training, validation, test rows	58, 15, 15	58, 15, 15	58, 15, 15
Linear regression, two features, MAE	0.74 kW	0.67 kWh/t	0.78 deg C
Linear regression plus load_x_ambient, MAE	0.75 kW	0.67 kWh/t	0.80 deg C
Linear regression plus load_squared, MAE	0.75 kW	0.67 kWh/t	0.78 deg C
Linear regression with load_pct only, MAE	3.12 kW	0.98 kWh/t	4.30 deg C
Ridge regression, degree 6 on load alone, alpha 1, MAE	3.05 kW	0.99 kWh/t	4.20 deg C
k-NN, k = 3, MAE	1.16 kW	0.63 kWh/t	1.58 deg C
Ridge regression, two features, alpha 1, MAE	0.84 kW	0.70 kWh/t	0.93 deg C
Decision tree depth 4, MAE	3.39 kW	1.20 kWh/t	4.79 deg C
Extended linear regression, intercept b_0 and weights b_1, b_2, b_3	4.8959, 0.3912, 0.6196, 0.0022	1.9344, 0.1074, 0.1276, 0.0005	6.3765, 0.6181, 0.8657, 0.0031
Ridge regression loss at alpha 1 (squared errors + L2 penalty)	50.4 + 110.6 = 161.1	35.7 + 7.7 = 43.4	65.3 + 262.5 = 327.8
k-NN worked example, nearest distance and prediction	d = 0.129, prediction 47.5 kW	d = 0.129, prediction 13.1 kWh/t	d = 0.129, prediction 70.0 deg C
Linear regression, cross-validated MAE (lowest to highest)	0.76 (0.74 to 0.77)	0.70 (0.57 to 0.78)	0.79 (0.75 to 0.87)
Lowest validation MAE among the four models	Linear regression	k-NN, k = 3	Linear regression

If things go wrong

- If the concepts run over, drop the scaling slide and say that ridge and k-NN need scaled features, as in Class 2. The formula slides and the worked-example tables are written for revision at home: read the formula aloud, point at one worked line and move on.
- If only forty minutes of lab remain, do steps 1, 2, 3 and 7, and set ridge regression and k-NN as practice in the rehearsal.
- If wifi fails in the room, the notebook runs unchanged in VS Code, as in Week 2.
- If several learners are stuck at the same cell, stop the room, run that cell on the projector and let them copy the fix.
- If the quiz must start early, cut the memo write-up and ask learners to finish it at home.

After the class

- Share the lab notebook and engineering_data.csv again so absent learners can follow.

- Release the rehearsal version of the project (not marked) and confirm the proposed sitting time for your cohort in writing.
- Note any question you could not answer and ask it before the Week 12 sitting.

Speaker script

Every slide carries its spoken script in the PowerPoint notes (open Presenter View). The scripts are first person and are meant to be read or paraphrased. Pauses are marked by questions.